

Robust cross-origin near-infrared prediction of soluble solids content in apples via a Beer–Lambert physics-informed prior

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ABSTRACT

Near-infrared (NIR) prediction of apple soluble solids content (SSC) often degrades across producing origins, and the physical mechanism is rarely quantified. We make three contributions. (i) Across 12 cross-origin Vis-NIR (591–1001 nm) scenarios, source-basis low-rank reconstruction residuals (PCA-10, NMF-3, SVD-3) amplify 1.1–4.9× on target domains and remain significant on held-out source samples (28/36 pairs, $p < 0.05$). (ii) A physics-inspired factorisation network, **Phys+BL**, encoding a Beer–Lambert decomposition in its latent space, attains 1.976 °Brix mean RMSE on 569 cross-origin scenarios, 23% better than a matched-budget CNN (95% CI [18.3, 26.6]). The gain is architecture-specific: the same loss on a plain CNN is harmful (−4.18%, $p \approx 10^{-6}$). (iii) Under matched-budget calibration with 20% target labels, slope/bias-corrected PLSR outperforms **Phys+BL** in every transfer type. The physical prior is therefore a diagnostic and architecture-probe tool, not an operational calibration champion when target labels exist.

1. Introduction

Non-destructive near-infrared (NIR) spectroscopic inspection of fruit quality is a mature commercial technology [1, 2]. Apple soluble solids content (SSC, i.e. sugar content) is a core metric for commercial grading [3, 4]; under controlled single-origin conditions, partial least squares regression (PLSR) models typically reach RPD > 2.0, meeting basic operational requirements [5, 6].

The problem is that such models often degrade significantly when deployed across origins. China’s major apple-producing regions span a vast area—Xinjiang, Shandong, Shaanxi and Gansu differ substantially in climate and soil—and fruits from different regions exhibit distinct distributions of cell wall density, intercellular air fraction, cuticular wax thickness and water activity gradients [7–9]. These tissue-structural differences alter the multiple-scattering baseline and effective optical path length of diffuse-reflectance NIR photons, introducing systematic spectral distribution shifts between origins. For example, applying a Shandong-calibrated model to Xinjiang apples typically yields RMSE > 3°Brix, more than twice the within-origin error [10]—a level that is essentially unacceptable in practice.

A variety of strategies have been proposed to bridge cross-origin shift, which can be grouped into three broad categories. *Spectral preprocessing* (MSC [11], SNV, derivatives, etc.) can partially normalise scattering differences, but these are essentially signal-level operations that do not exploit the physical structure underlying the shift [12, 13]. *Statistical domain adaptation* (TCA [14], CORAL [15], MMD-penalised networks [16, 17], ADDA [18], di-PLS [19]) aligns feature distributions, but in the absence of physical constraints can easily induce negative transfer when the spectral manifold is curved [20, 21]. For NIR spectroscopy, regression-oriented domain adaptation under a BL view has been explored [22, 23]; these methods primarily introduce domain-invariance constraints within a shallow PLS framework, and are complementary to our route of encoding BL structure directly into the latent space of a deep network. *Empirical calibration transfer* methods (DS and PDS [24, 25]) typically require paired calibration samples in the target domain, making them difficult to apply in zero-label deployment settings [7, 26]. Recent multi-output CNNs have been proposed for simultaneous fruit-quality prediction [27], generative adversarial dual-channel networks for cross-origin food spectroscopy [28], and broader CNN-based postharvest reviews [29]; these advances remain orthogonal to the structural-prior strategy explored here.

Existing methods mitigate cross-origin shift primarily through spectral preprocessing or feature-distribution

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